

NexRay AI

Medical Assistant Platform • Clinical Decision Support Report

25 July 2026 | 04:30 PM

Session ID: 29	Modules Used: Symptom Checker • Refined Diagnosis	Date: 25 July 2026 04:30 PM
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Symptom Analysis

Possible Condition	Confidence
Typhoid Fever (Enteric Fever)	88%
Malaria with complications	62%
Acute Bacterial Gastroenteritis (non-typhoidal)	48%

■ **Urgency Level: Urgent — Same day attention required**

Recommended Tests

- Blood culture (gold standard for typhoid confirmation)
- Widal test (agglutination test for Salmonella typhi and paratyphi)
- Rapid malaria test/blood smear microscopy
- Complete blood count (CBC)
- Liver function tests and renal function tests
- Abdominal ultrasound (to assess for splenomegaly and rule out other acute abdominal pathology)

Suggested Treatment

- Initiate empiric broad-spectrum antibiotics (fluoroquinolone or third-generation cephalosporin) pending culture results, as typhoid is most likely
- Supportive care: IV fluids for rehydration and electrolyte management
- Antipyretics for fever management (paracetamol or ibuprofen)
- Monitor for complications such as perforation, encephalopathy, or myocarditis

Next Steps

- Obtain blood cultures immediately before starting antibiotics if possible
- Perform Widal test and rapid malaria screening
- Monitor vital signs closely and assess for signs of typhoid complications
- If malaria test is positive, initiate appropriate antimalarial therapy concurrently
- Review antibiotic choice once culture sensitivities are available

Summary: A 5-day sustained fever with abdominal pain, loss of appetite, weakness, constipation, and characteristic rose-coloured spots strongly suggests Typhoid Fever, which is endemic in West Africa. Malaria must be ruled out given its high prevalence in the region. Blood cultures and Widal testing are critical for confirmation. Empiric antibiotic therapy should be initiated urgently while awaiting confirmatory results. Close monitoring for complications is essential.

Refined Diagnosis — Based on Test Results

Final Diagnosis: Confirmed Typhoid Fever caused by Salmonella typhi with hepatic involvement and splenomegaly.

■ Confirmed Conditions

- Typhoid Fever (Enteric Fever)** (Confirmed) — Blood culture positive for Salmonella typhi. Widal test significant titres. Splenomegaly and leukopenia confirm typhoid.

■ Ruled Out

- Malaria with complications** — Rapid malaria test negative.
- Acute Bacterial Gastroenteritis** — Blood culture confirmed Salmonella typhi ruling out non-typhoidal gastroenteritis.

■ **Urgency Level: Urgent — Same day attention required**

Updated Treatment

- Ciprofloxacin 500mg BD or ceftriaxone 2g IV daily for 7-14 days
- Oral rehydration or IV fluids for hydration
- Paracetamol for fever — avoid NSAIDs
- Monitor for complications including perforation and encephalopathy

Next Steps

- Antibiotic susceptibility testing from blood culture
- Daily assessment for complications
- Repeat blood cultures at 48-72 hours if fever persists

Summary: Typhoid Fever confirmed by positive blood culture. Malaria excluded. Antibiotic therapy initiated with close monitoring.

■ **DISCLAIMER:** This report is generated by NexRay AI and is intended solely as a clinical decision-support tool. All findings, conditions, treatments, and recommendations are AI-generated suggestions and do not constitute a formal medical diagnosis or prescription. The clinical judgment of the attending medical professional must be applied before any action is taken.